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1 Writing Conventions

In general I use British English.

1.1 Logical rules

Each item in a list will end with a point and start capitalized. If I list something where the order matters like:

1. Step one.
2. Step two.
3. Step three.

and for equivalent properties I use letters: For example, a Morse-Smale function f on a Riemannian manifold (M, g) satisfies:

- (a) f is smooth.
- (b) The critical points of f are non degenerate.
- (c) The (un-) stable manifolds intersect transversal.

1.2 Structure

In this thesis i have the following text-types

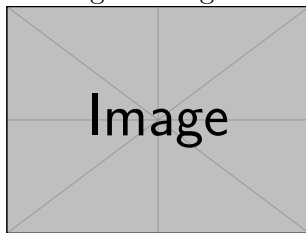
- (a) Free text that is not in any container.
- (b) The theorem container.
- (c) The lemma container.
- (d) The corollary container.
- (e) The proposition container.
- (f) The prove container.

Free Text is the main narrative of the thesis. It includes explanations, motivations, intuitions, examples, informal remarks, and transitions. Free text connects the formal content and helps guide the reader through the logical structure of the work.

Theorem 1.2.1 (An Example). *Used for major, central mathematical statements that are either original contributions or critical results from the literature. Theorems typically capture the most important claims and are often proved in detail.*

Proof Strategy:

Here I give the geometric idea of the proof and explain the needed steps.

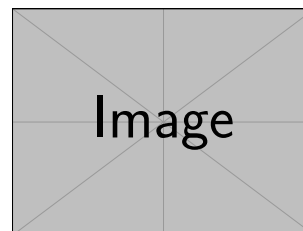


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Lemma 1.2.2 (An Example). *Used for technical results that support the proof of*

theorems or other key statements. Lemmas are usually not of independent interest but are essential building blocks within a logical argument.

Corollary 1.2.3 (An Example). Used for statements that follow immediately and straightforwardly from a previously proven theorem or lemma. Corollaries are usually included to highlight direct consequences of major results.

Proposition 1.2.4 (An Example). *Used for results that are mathematically relevant but not as central as theorems. Propositions often encapsulate useful facts, structural properties, or classifications that are too substantial for a lemma but not significant enough for a theorem.*

Proof. This container is pretty obvious □

1.3 Mathematical Conventions

Elements in a vector bundle will have greek letters. For example $\xi \in TM$. For restrictions I will use the notation $f|_A$ that is written as :

`\restr{f}{A}`.

In the realm of differential geometrie i will **always** use lower indices to denote components. I will **not** use the einstein summation, since it really appears and I never really deal with cumbersome computations (in this context).

1.4 Notation

Symbol	Description	Context
A^t	The transpose of the Matrix A	Linear algebra
M	Smooth manifold	Differential geometry
$\dim M$	Dimension of manifold M	Differential geometry
TM	Tangent bundle of M	Differential geometry
$T_p M$	Tangent space at point p	Differential geometry
g	Riemannian metric	Riemannian geometry
$\text{Hess}_p(f)$	Hessian of f in p	Morse theory
$M_p(f)$	The matrix correspodng to the Hessian. The used chart is suppressed	Morse theory
$\text{Crit}_k(f)$	Set of critical points of index k	Morse theory
λ_p or λ_p^f	Index of the critical point p	Morse theory

2 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of interest for mathematicians and physicists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 2.0.1 (Topological Manifold). A second countable Hausdorff space M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p , an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : U \rightarrow V$ that is a homeomorphism. We call the map $\varphi : U \rightarrow V$ a **chart around p on M** and φ^{-1} a **local coordinate system around p on M** .

Definition 2.0.2 (Differentiable Manifold). Let M be a topological manifold of dimension m .

1. A **differentiable atlas of class $r \in \mathbb{N} \cup \{\infty\}$** is a family of charts $\mathfrak{A} = (\varphi_i : U_i \rightarrow V_i)_{i \in I}$ such that
 - a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M .
 - b) For every pair $(i, j) \in I^2$ the **transition function**:

$$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\rightarrow \varphi_i(U_i \cap U_j) \\ x &\mapsto (\varphi_i \circ \varphi_j^{-1})(x) \end{aligned}$$

is differentiable of class r .

We call such an atlas a C^r -atlas.

2. Two C^r -atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r -atlas.

A **differentiable structure of class r** on M is an equivalence class c of C^r -atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 2.0.3. Every transition functions φ_{ij} $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$.

Definition 2.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \rightarrow \mathbb{R}$$

differentiable of class r , if for any one (and hence for all) $(\varphi_i)_{i \in I} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r . For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f \in C(U, \mathbb{R}) \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 2.0.5. Let (M, c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}$ -algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$ -algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \text{res}_V^U : \mathcal{E}(U) &\rightarrow \mathcal{E}(V) \\ f &\mapsto f|_V . \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. $\square$

Definition 2.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p , if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p . We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{ open}, p \in U} \mathcal{E}(U) \right) / \sim_p .$$

For the set of germs and call it the **stalk in** p . Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 2.0.7. For a smooth manifold (M, c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$ -algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \text{eval}_p : \mathcal{E}_p(M) &\rightarrow \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\text{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\text{eval}_p)$. $\square$

Definition 2.0.8. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$ -linear map $\delta : \mathcal{E}_p(M) \rightarrow \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p)g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f, g \in \mathcal{E}_p(M) .$$

We call $\text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R}) .$$

Corollary 2.0.9. Let (M, c) be a smooth manifold and $\varphi : U \rightarrow V$ be a chart around p with $x_0 = \varphi(p)$ ($\varphi \in \mathfrak{A} \in c$). Then

$$\xi = \frac{\partial}{\partial x_j} \Big|_p : \mathcal{E}_p(M) \rightarrow \mathbb{R}$$

$$f_p \mapsto \xi(f_p) = \frac{\partial}{\partial x_j} \Big|_{x_0} (f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p \right)$$

defines a basis of TM_p . Hence, the dimension of TM_p is m .

Definition 2.0.10. For a smooth manifold (M, c) the sum $\sum_p TM_p$ comes with a natural projection

$$\pi : TM \rightarrow M$$

$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \rightarrow V$

$$\frac{\partial}{\partial x_j} : U \rightarrow \pi^{-1}(U)$$

$$p \mapsto \frac{\partial}{\partial x_j} \Big|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m.$$

We can induce a topology on TM such that all those trivializations are continuous. This then gives an atlas for TM such that we have a $2m$ -dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \rightarrow \mathbb{R}^m \times \mathbb{R}^m; x \mapsto (\pi(x), q_{\varphi \circ \pi}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p)$ that depends on the chart φ_i . In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m . We call TM the **tangent bundle**.

Definition 2.0.11 (The Derivative). Let (M, c) and (M', c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \rightarrow M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \rightarrow V'$$

are smooth. A given smooth function induces a smooth function between the Tangent bundles as follows:

$$Df : TM \rightarrow TM', Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_p M$, $g_p \in E_p(M')$.

Corollary 2.0.12. Let $f : M \rightarrow \mathbb{R}$ be smooth and $\varphi : U \rightarrow V$ be a chart. Then we can interpret df as a one-form. To be precise assume q to be the coordinate function $T\mathbb{R} \rightarrow \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ df(v) = v \cdot f$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notation so vector fields can take functions as an input.

Proof. We prove this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M, \mathbb{R})$ vectorspace. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$df\left(\frac{\partial}{\partial x_i}\Big|_p\right)(g_p) = \frac{\partial}{\partial x_i}\Big|_p(g \circ f)_p = \frac{\partial}{\partial x_i}\Big|_{x_0}(g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\Big|_p(g_p) \cdot \frac{\partial}{\partial x_i}\Big|_p(f_p)$$

Hence, $df(v) = v(f)\frac{\partial}{\partial x}$ letting us conclude the statement. $\square$

Definition 2.0.13 (A Metric). Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual Tangent bundle with itself is a **riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_p M$ $g_p(v_p, w_p) = 0 \ \forall w_p \in T_p M$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p) \ \forall v_p, w_p \in T_p M$.
3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0 \ \forall v_p$ and vanishes only for $v_p = 0$.

We call a tuple (M, g) a **riemannian manifold**.

Definition 2.0.14. Let M, N, L be smooth manifolds and $f : M \times N \rightarrow M' \times N'$. Let $X \in \Gamma(TM)$ and $Y \in \Gamma(TN)$. We define $\tilde{X} \in \Gamma(T(M \times N))$ as follows. Let $f_p \in \mathcal{E}_p(M, N)$

$$\tilde{X}(f_p) = X((f \circ i_M)_{p_M}) \quad \text{where } p_M = \pi_1(p).$$

Definition 2.0.15 (The Gradient). Assume that (M, g) is a smooth manifold together with a riemannian metric. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\text{grad}(f) \in \Gamma(TM)$ such that for any vector field V :

$$q \circ df(V) = g(\text{grad}(f), V).$$

Here q denotes the canonical coordinate function $T\mathbb{R} \rightarrow \mathbb{R}$.

Definition 2.0.16. Let (M, g) be a riemannian manifold and $\varphi : U \rightarrow V$ be a chart. We define the smooth functions $g_{ij} : U \rightarrow \mathbb{R}$ to be :

$$g_{ij} = g\left(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j}\right).$$

Then if two vectorfields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v, w) = \sum_{ij} g_{ij} v^i w^j : U \rightarrow \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via crammers rule) to get smooth functinos

$$g^{ij} : U \rightarrow \mathbb{R}$$

$$p \mapsto g^{ij}(p) = ((g_{kl}(p))_{kl}^{-1})_{ij}$$

Lemma 2.0.17. *Let $\varphi : U \rightarrow V$ be a chart. Then the gradient has the local form:*

$$\text{grad}(f) = \sum_{i,j} g^{ij} \left(\frac{\partial}{\partial x_i}(f) \right) \frac{\partial}{\partial x_j} = \sum_{i,j} g^{ij} \frac{\partial f}{\partial x_i} \frac{\partial}{\partial x_j}.$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum w^i \frac{\partial}{\partial x_i}$. Then $g(v, w) = \sum_{i,j} g_{ij} v^i w^j$. Suppose that $\text{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \rightarrow \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ df \left(\frac{\partial}{\partial x_j} \right) = q \circ \frac{\partial f}{\partial x_j} = g(\text{grad}(f), \frac{\partial}{\partial x_j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$(G^1, \dots, G^m)(g_{ij}) = \left(\frac{\partial}{\partial x_1}(f), \dots, \frac{\partial}{\partial x_m}(f) \right)$$

$$\Leftrightarrow (G^1, \dots, G^m) = \left(\frac{\partial}{\partial x_1}(f), \dots, \frac{\partial}{\partial x_m}(f) \right) (g^{ij}).$$

But then we can conclude that $G^j = \sum_i g^{ij} \frac{\partial}{\partial x_i}(f)$.

□

Definition 2.0.18 (A Dynamical System). Let M be a smooth manifold. A **dynamical system on M** is given by a smooth map $\varphi : \Omega \rightarrow M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t, p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t, p))) = \varphi(s + t, p)$$

3. For all p we require $\varphi(0, p) = p$.

We will write $\varphi : t(p) := \varphi(t, p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 2.0.19. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \rightarrow \text{Diff}, \quad t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t .

Definition 2.0.20 (Vector Field of a Dynamical System). Let φ be a dynamical system on a smooth manifold M . Then we call the mapping

$$p \mapsto X(p) := \left. \frac{d}{dt} \right|_{t=0} \varphi_t(p) \in T_p M$$

the **associated vector field**. This is in fact a smooth vector field.

Whilst differentiating leads to a smooth vector field, we can also go the other direction by locally integrating a vector field to get a flow. To be precise: Those concepts are inverse to one another or in other words

Theorem 2.0.21. *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

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3 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [banyaga2004lectures]. In Morse Theory, we will always deal with a smooth function $f : M \rightarrow \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 3.0.1 (Kritical Points). We call a point $p \in M$ **critical**, if $df|_p$ vanishes meaning that

$$df|_p : T_p M \rightarrow T_{f(p)} \mathbb{R}$$

is trivial. By $\text{Crit}(f)$ we denote the **set of critical points**.

Definition 3.0.2 (The Hessian). Let $f : M \rightarrow \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \text{Crit}(f)$ is a symmetric bilinear function

$$\text{Hess}_p(f) : T_p M \times T_p M \rightarrow \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_p M$ be two tangend vectors. Now choos two vectorfields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\text{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))(p)$$

Compare 2.0.12 for the notation.

Definition 3.0.3. A smooth function $f : M \rightarrow \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a Symmetric bilinear and non-degenerate Map. For such a map there exists the following theorem:

Theorem 3.0.4 (Sylvesters Law of Inertia). *Let $A \in \text{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \text{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k', l = l'$ and $\text{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \text{rank}(T^t \circ A \circ T) = \text{rank}(A) = \text{rank}(T'^t \circ A \circ T') = k' + l'. \quad (1)$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max \{ \dim(U) \mid U \subseteq \mathbb{R}^n \text{ subspace such that } x^t A x > 0 \ \forall x \in U \setminus \{0\} \}$$

So we start by showing „ $\leq$ “: Denote the first k colomus of T with $x_1, \dots, x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda_i x_i$ we have that by bilinearity

$$x^t A x = \sum_{i=1}^k \lambda_i x_i^t A x = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^t A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0. \quad (2)$$

This concludes the first inequality.

Now let U be any k -dimensional subspace such that for all non zero $x \in U$ $x^t A x > 0$. By a calculation analog to the one above we have that for any $x \in W := \text{span}(x_{k+1}, \dots, x_n)$ the number $x^t A x$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k. \quad (3)$$

This is the last inequality proving the statement. $\square$

Corollary 3.0.5. Let $\varphi : U \rightarrow V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corresponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)_{i,j}$$

Proof. This is just an easy calculation, since $\text{Hess}_p(f) \left(\frac{\partial}{\partial x_i} \Big|_p, \frac{\partial}{\partial x_j} \Big|_p \right)$ is given by

$$\left(\frac{\partial}{\partial x_i} \Big|_p \left(\frac{\partial}{\partial x_j} (f) \right) \Big|_p \right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)$$

$\square$

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \rightarrow V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function. If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transforms from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = d\varphi_{12}^t \cdot M_p^2(f) \cdot d\varphi_{12}$$

Hence we can define the following:

Definition 3.0.6. By Sylvesters Law of inertia, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p , denoted with λ_p . If we cannot suppress the function without loss of clarity we will write λ_p^f . We denote the set of all critical points of index k by $\text{Crit}_k(f)$.

Theorem 3.0.7 (Morse Lemma). *Let p be a critical point of index k of the Morse funktion $f : M \rightarrow \mathbb{R}$. There exists a centered chart $\varphi : U \rightarrow \mathbb{R}^m$ around p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^k -x_i^2 + \sum_{i=k+1}^m x_i^2$$

Proof. Compar for example Lemma 3.11 in [banyaga2004lectures]. □

hier noch
mehr
Quellen!

Definition 3.0.8. Let $\varphi_t : M \rightarrow M$ be the local one-parameter group of diffeomorphisms generated by $-\text{grad}(f)$, i.e. if $e : \mathbb{R} \rightarrow T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{d}{dt} \varphi_t(x) \circ e = -\text{grad}(f)(\varphi_t(x)), \quad \text{and} \\ \varphi_0(x) = x.$$

We call that group the **gradient vector field** of f with respect to the gradient g . The integral curve $\gamma_x : (a, b) \rightarrow M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a **gradient flow line**.

Next we will deepen our **local** understanding of said vector field. Notice how the Morse Lemma is a statement about smooth manifolds-no metric needed nor mentioned. In fact there are many Morse charts and we can find one, where the riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 3.0.9. *Assume that $\psi : U \rightarrow V$ is a chart and $L : \mathbb{R}^m \rightarrow \mathbb{R}^m$ a linear function. Denote the local coordinate maps of the Tangent bundle in a point p that comes from a basis in said point induced from a chart χ by q_χ . Then the diagram*

$$\begin{array}{ccc} T_p M & \xrightarrow{q_\psi|_p} & \mathbb{R}^m \\ & \searrow q_{L \circ \psi}|_p & \downarrow L \\ & & \mathbb{R}^m \end{array}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi \circ L}|_p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x}_0$, $(v_1, \dots, v_m) = v \in \mathbb{R}^m$ and denote the basis of $T_p M$ coming from

the chart ψ by $\left(\frac{\partial}{\partial x_i}\bigg|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\bigg|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
 \left[(q_{\psi \circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\bigg|_p(f_p) \\
 &= \sum_i v_i \frac{\partial}{\partial x_i}\bigg|_{\tilde{x}_0}(f \circ \psi^{-1} \circ L^{-1}) \\
 &= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\bigg|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\bigg|_{\tilde{x}_0} \\
 &= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\bigg|_p(f_p) \cdot \lambda_{ji} \\
 &= \sum_{i,j} v_i q_{\psi}\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
 &= \left[q_{\psi}\big|_p^{-1}\left(\sum_{i,j} v_i \lambda_{ji} e^j\right)\right](f_p) \\
 &= \left[q_{\psi}\big|_p^{-1}(L^{-1}(v))\right](f_p)
 \end{aligned}$$

This concludes the proof. $\square$

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 3.0.10 (Simultaneous Diagonalisation of Quadratic Forms). *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on $T_p M$. Then there exists a Morse chart ψ around p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a diagonal matrix $\text{diag}(\mu_1, \dots, \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precise this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_{\psi}\big|_p \oplus q_{\psi}\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda Gv$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and corresponding eigenvectors $v_1, \dots, v_m$, which can be chosen to be orthogonal with

respect to the bilinear form defined by G , such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (H v_j)^T v_i = \lambda_j (G v_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i).$$

Hence we have the equality

$$\lambda_j (G v_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1 | \dots | v_m]$ be the matrix whose columns are these G -orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z . In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^m z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H(Lz) = z^T L^T H L z \end{aligned}$$

Since $H v_i = \lambda_i G v_i$, we have $L^T H L = \text{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m-l$ positive). This is true to Sylvester's law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\text{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \text{diag}(\mu_1, \dots, \mu_m)$ with $\mu_j > 0 \ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y)) = \sum_{i=1}^l -y_i^2 + \sum_{i=l+1}^m y_i^2$$

Furthermore, we want to inspect the metric $g|_p : T_p M^2 \rightarrow \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$ -induced basis $\left(\frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p \right)$. By the lemma 3.0.9

$$g(q_{L^{-1} \circ \psi} \Big|_p^{-1}, q_{L^{-1} \circ \psi} \Big|_p^{-1})(v, w) = g(q_\psi \Big|_p^{-1}, q_\psi \Big|_p^{-1})(Lv, Lw) = (Lv)^T G L w = v^T L^T G L w$$

□

Definition 3.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulas around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 3.0.12 (Jacobian of the Gradient). The gradient is a section into the tangent bundle, $\text{grad}(f) : M \rightarrow TM$. Let $\psi : M \supseteq U \rightarrow V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_p M$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\text{grad}(f))_\psi(x) := J(q_\psi \circ \text{grad}(f) \circ \psi^{-1})(x) = \left(\frac{\partial(q_\psi \circ \text{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x) \right)_{ij}$$

Lemma 3.0.13. *Let ψ be a coordinate system around a critical point p . I.e. $\psi : p \in U \rightarrow V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of $T_x M$. Then the Jacobian of the gradient reads*

$$J(\text{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right)_{ij}. \quad (4)$$

Proof. Let q_ψ denote the coordinate funktion $TU \rightarrow \mathbb{R}^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \text{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k} \right)$$

Hence, we can differentiate:

$$\begin{aligned} \frac{\partial}{\partial x} q_\psi \circ \text{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k} \right)_{ij} \\ &= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right] \right)_{ij}. \end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f) \circ \psi^{-1}\big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right] \right)_{ij}.$$

□

Now that we now how the gradient locally looks like it is reasonable to assume, that the flow locally looks like its exponential:

Lemma 3.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \rightarrow M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ such that $\varphi_t(p) \in U$. Then the linear map $d\varphi_t|_p : T_p M \rightarrow T_{\varphi_t(p)} M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) = \exp \left(- \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} \right)_{ij} \cdot t \right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \rightarrow U$. For fixed x we can calculate

$$q_\psi \frac{d}{dt} \varphi_t(\psi^{-1}(x)) = -q_\psi \circ \text{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned} \frac{\partial}{\partial t} \frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x} \frac{\partial}{\partial t} \psi \circ \varphi_t(\psi^{-1}(x)) \\ &= \frac{\partial}{\partial x} q_\psi \frac{d}{dt} \varphi_t(\psi^{-1}(x)) \\ &= -\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f)(\varphi_t(\psi^{-1}(x))) \\ &= -\frac{\partial}{\partial x} q_\psi \circ \text{grad}(f)(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))) \\ &= -\frac{\partial}{\partial x} (q_\psi \circ \text{grad}(f) \circ \psi^{-1}) \frac{\partial}{\partial x} (\psi \circ \varphi_t(\psi^{-1}(x))) \end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x} \psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) = \exp \left(-\frac{\partial}{\partial x} (q_\psi \circ \text{grad}(f) \circ \psi^{-1}) \cdot t \right).$$

The preceding lemma now finishes the proof. □

Hence in a refined Morse chart we have:

Remark 3.0.15. Let ψ be a refined Morse chart, and hence:

- (a) $g^{ik}(0) = \delta_{ik} \mu_k$ with $\mu_k > 0$ for any k .
- (b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}
 \frac{\partial}{\partial x} \psi \circ \varphi_t(\psi^{-1}(x)) &= \exp(-J(\text{grad}(f))(p) \cdot t) \\
 &= \exp \left(- \sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k} (0) \underbrace{\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}}_{=s_k 2\delta_{jk}} \cdot t \right) \\
 &= \text{diag} \left(e^{2\mu_1 t}, \dots, e^{2\mu_{\lambda_p} t}, -e^{2\mu_{\lambda_p+1} t}, \dots, -e^{2\mu_m t} \right)
 \end{aligned}$$

4 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

4.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 4.1.1 (Compact invariant isolated subset). For a flow $\varphi_t : M \rightarrow M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t \in \mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 4.1.2 (Index pairs). Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of S**, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (N \setminus L)$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in N**. Here $\varphi_{[0,t]}(x) := \{\varphi_t(x) \mid t \in [0, t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S . Due to the third property we call L the **exit set**.

Definition 4.1.3 (Regular index pairs). We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 4.1.4 (Homotopy equivalence of index pairs). *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S . Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces.*

Remark 4.1.5 (Critical points as isolated compact invariant subsets). Let $f : M \rightarrow \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M, g) . Let $q \in \text{Crit}_k(f)$. Around q we have coordinate charts $\varphi : U \rightarrow T_q M$ (after identifying q_i and ∂q_i) where $\varphi(W(q \rightarrow) \cap U) \subseteq T_q^u M$ and $\varphi(W(\rightarrow q) \cap U) \subseteq T_q^s M$. With this we can define the balls:

$$\begin{aligned} D_\varepsilon^s &= \{v \in T_q^s M \mid \|v\| \leq \varepsilon\}, \\ D_\varepsilon^u &= \{v \in T_q^u M \mid \|v\| \leq \varepsilon\}. \end{aligned}$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D_\varepsilon^u, \partial D_\varepsilon^u)$. Now we now the Kgroup of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i - k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \quad (5)$$

Notice that for $k = 0$ we need to consider $\partial D^k = \emptyset$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k :

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) = \bigoplus_{\text{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \quad (6)$$

Furthermore, this isomorphism can be made canonically, by using orientations: because we get canonically induced generators by orientations of the unstable manifolds.

Example 4.1.6 (A map of homology groups). Let $f : M \rightarrow \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g) . Let $q \in \text{Crit}_k(f)$ and $p \in \text{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \rightarrow q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$ depicted in figure ??, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0) is a pair for p . Assuming we have regular index pairs (N_q, L_q) and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \rightarrow p) : K^{k-1}(N_p, L_p) \rightarrow K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{triple}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \text{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \rightarrow \bigoplus_{q \in \text{Crit}_k(f)} K^k(N_q, L_q) \quad (7)$$

Definition 4.1.7 (Regular index pairs and a filtration on M). As always let $f : M \rightarrow \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m . Let $\varphi_t : M \rightarrow M$ be the flow given by $-\text{grad} f$. For $0 \leq j \leq k \leq m$ define:

$$W(k, j) = \bigcup_{j \leq \lambda_p \leq \lambda_q \leq k} W(q \rightarrow p).$$

We know that this space is compact. Assume that N is a compact neighbourhood of

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$W(k, j)$, such that $\text{Crit}(f) \cap N = \text{Crit}(f) \cap W(k, j)$, meaning that N does not contain any new critical points. Then the biggest invariant subspace from definition 4.1.2 is

$$I(N) := \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} = W(j, k).$$

This follows from every gradient flow line starting and ending in a critical point. Therefore, $W(k, j)$ is an isolated compact invariant set. By a corollary of the lambda lemma we know that

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$$\begin{aligned} W_j^s &:= \bigcup_{j \leq \lambda_p} W(\rightarrow p) \\ W_j^u &:= \bigcup_{\lambda_p \leq j} W(p \rightarrow) \end{aligned}$$

for all $j = 0, \dots, m$ are compact as they are finite unions of compact sets. Now we define $N_m := M$ and choose a cofibered compact neighbourhood N_{m-1} of $W_{m-1}^{p \rightarrow}$ that is positively invariant and satisfies $N_{m-1} \cap W_m^{\rightarrow p} = \emptyset$. One could for example define:

$$N_{m-1} := N_m \setminus \left(\bigcup_{q \in \text{Crit}_m(f)} \overset{\circ}{N}_q \right),$$

where N_q is taken from example 4.1.5. Once we check that N_{m-1} is an exit set with respect to N_m we can conclude that (N_m, N_{m-1}) is a regular index pair for $\text{Crit}_m(f)$. The first statement is clear, since every gradient line starting in $N_m \setminus N_{m-1}$ has to pass through N_{m-1} , as they go towards other critical points. This tells us that they have to leave $N_m \setminus N_{m-1}$, which is enclosed by N_{m-1} . The second thing to check can be done by showing that there is a neighbourhood $U \subset N_m$ of N_{m-1} that deforms to N_{m-1} . For this notice that each N_q with $\lambda_q = m$ is homeomorphic to D^m . Now choose a open neighbourhood U_q in the disc of its boundary. By definition this retracts to the boundary with a retraction induced by the flow. And finally define $U := N_{m-1} \cup_{q \in \text{Crit}_m(f)} U_q$. This retracts to N_{m-1} telling us that our index pair is indeed regular.

Now we want to inductively define a filtration: For this we choose a compact cofibered neighborhood N_{m-2} of W_{m-2}^u that is positively invariant in N_{m-1} and has an empty intersection with $W_{m-1}^{\rightarrow p}$. This can be done similar to the construction of N_{m-1} but instead of cutting out neighbourhoods if $q \in \text{Crit}_m(f)$, we can cut out tubular neighborhoods of W_{m-1}^s . By iterating this process we get a filtration:

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \dots \subset N_{m-1} \subset N_m = M, \quad (8)$$

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Salamon

such that (N_k, N_{j-1}) is a regular index pair for $W(k, j)$ for all $0 \leq j \leq k \leq m$.

Definition 4.1.8. Let M be a smooth manifold with morse data and

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \dots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 4.1.7. Then we define the Groups

$$E_r^{p,q}(M) := \frac{\ker [K^{p+q}(N_{p+r-1}, N_{p-r}) \rightarrow K^{p+q}(N_{p-1}, N_{p-r})]}{\ker [K^{p+q}(N_{p+r-1}, N_{p-r}) \rightarrow K^{p+q}(N_p, N_{p-r})]}.$$

To define a boundary operator, we proceed as follows: The following diagram commutes (by naturality of the boundary in the triple sequence).

$$\begin{array}{ccccc} K^{p+q}(N_{p+r-1}, N_{p-r}) & \xrightarrow{\alpha} & K^{p+q}(N_p, N_{p-r}) & & \\ \downarrow \delta_{triple}^* & & \downarrow \delta_{triple}^* & & \\ K^{p+q+1}(N_{p+2r-1}, N_{p+r-1}) & \xrightarrow{\beta} & K^{p+q+1}(N_{p+2r-1}, N_p) & \longrightarrow & K^{p+q+1}(N_{p+r-1}, N_p) \end{array}$$

Hence, $\beta \circ \delta_{triple}^*$ factors over $K^{p+q}(N_{p+r-1}, N_{p-r})/\ker \alpha$ and because the bottom row is exact (by the triple sequence) we have get a map

$$K^{p+q}(N_{p+r-1}, N_{p-r})/\ker \alpha \rightarrow \ker [K_{p+q+1}(N_{p+2r-1}, N_p) \rightarrow K^{p+q+1}(N_{p+r-1}, N_p)]$$

Now since $K^{p+q}(N_{p+r-1}, N_{p-r}) \supseteq \ker [K^{p+q}(N_{p+r-1}, N_{p-r}) \rightarrow K^{p+q}(N_{p-1}, N_{p-r})]$ we can restrict the domain and furthermore compose with the quotient map to get a well-defined map

$$\begin{aligned} d_r^{p,q} : E_r^{p,q} &\rightarrow E_r^{p+r, q-r+1} \\ [x] &\mapsto [\beta \circ \delta_{triple}^*(x)] \end{aligned}$$

Corollary 4.1.9 (The first page). We start by analysing the first page

$$\begin{aligned} E_1^{p,q}(M) &:= \frac{\ker [K^{p+q}(N_p, N_{p-1}) \rightarrow K^{p+q}(N_{p-1}, N_{p-1})]}{\ker [\underbrace{K^{p+q}(N_p, N_{p-1}) \rightarrow K^{p+q}(N_p, N_{p-1})}_{\text{id}}]} \cong K^{p+q}(N_p, N_{p-1}) \\ &= \tilde{K}^{p+q}(N_p/N_{p-1}) \end{aligned}$$

Now since (N_p, N_{p-1}) is a regular index pair for $\text{Crit}_p(f)$ we have a flow induced homotopy equivalence:

$$N_p/N_{p-1} \simeq \left(\bigcup_{w \in \text{Crit}_p(f)} D_\varepsilon^u(w) \right) / \left(\bigcup_{w \in \text{Crit}_p(f)} \partial D_\varepsilon^u(w) \right) \cong \bigvee_{w \in \text{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D_\varepsilon^u(w) := \varphi_w^{-1}(\{v \in T_w^u M \mid \|v\| \leq \varepsilon\}),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$\begin{aligned} E_1^{p,q} &\cong \tilde{K}^{p+q}(N_p/N_{p-1}) \\ &\cong \bigoplus_{w \in \text{Crit}_p(f)} \tilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \tilde{K}^q(pt)) \end{aligned}$$

In sum we have the chain of horizontal isomorphisms:

$$\begin{array}{ccccccc}
E_1^{p,q} & \xrightarrow{\alpha} & \tilde{K}^{p+q}(N_p/N_{p-1}) & \xrightarrow{\beta} & \oplus_{w \in \text{Crit}_p(f)} \tilde{K}^q(pt) & \longleftarrow & C^p(M, \mathfrak{A}, \tilde{K}^q(pt)) \\
\downarrow d_1^{p,q} & & \downarrow & & \downarrow & & \downarrow \\
E_1^{p+1,q} & \xrightarrow{\alpha'} & \tilde{K}^{p+1+q}(N_{p+1}/N_p) & \xrightarrow{\beta'} & \oplus_{w \in \text{Crit}_{p+1}(f)} \tilde{K}^q(pt) & \longleftarrow & C^{p+1}(M, \mathfrak{A}, \tilde{K}^q(pt))
\end{array}$$

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology?

The Main statement now is that the connecting homomorphism of K-Theory and of the long exact sequence given by a special Filtration in Morse Theory are comparable:

Theorem 4.1.10. *Given the filtration $N_{-1} \subseteq N_0 \subseteq \dots \subseteq N_{m-1} \subseteq N_m = M$ from (8). The following diagram commutes:*

$$\begin{array}{ccc}
C^k(M, \mathfrak{A}, \tilde{K}^l(pt)) & \xrightarrow{\partial^p} & C^{k+1}(M, \mathfrak{A}, \tilde{K}^l(pt)) \\
\downarrow & & \downarrow \\
\bigoplus_{p \in \text{Crit}_k(f)} \tilde{K}^{k+l}(N_p, L_p) & \xrightarrow{\Delta_{k+l}} & \bigoplus_{q \in \text{Crit}_{k+1}(f)} \tilde{K}^{k+l+1}(N_q, L_q) \\
\downarrow & & \downarrow \\
K^{k+l}(N_k, N_{k-1}) & \xrightarrow{\delta_{triple}} & K^{p+l+1}(N_{k+1}, N_k)
\end{array}$$

Proof Strategy:

We will first build up a diagram that encapsulates the connecting homomorphism from K-theory, which was build as follows: Assume that (X, Y, Z) is a topological pair. Then the first map is induced from the kollaps: $Y \rightarrow Y/Z$. Afterwards we have the connecting homomorphism of the pair:

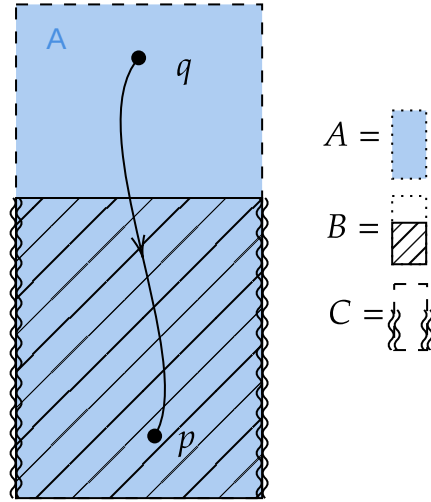
Then the K-groups of the pointed spaces X/X and $X \cup CY$, where the latter denotes a cone above Y agree. From the latter we can quotient out X , and $X \cup CY/X \cong SY$. Hence we have the triple connecting homomorphism as the composition:

$$K^{-1}(Y, Z) \rightarrow K^{-1}(Y) = K(SY) \cong K(X \cup CY/X) \rightarrow K(X \cup CY) \cong K(X/Y)$$

The triple connecting homomorphism for lower K-Groups is defined exactly the same via the identification $K^{-n}(X, Y) = K(S^n \vee (X, Y))$. For higher K-Groups we apply the natural Bott homomorphism. We first proof it here for the lower groups and assume k, l to be negative. Furthermore we assume for now that $l = 0$. If l is even, we can apply the Bott isomorphism and else the groups are 0. Now we choose

a triple (A, B, C) in M such that:

- (a) (A, B) is a regular index pair for a critical point $q \in \text{Crit}_{k+1}$.
- (b) (B, C) is a regular index pair for a critical point $p \in \text{Crit}_k$.



From the perspective of K-theory a regular index pair for a critical point is the same as a sphere with the dimension being the index of said critical point. To see this just use a small index pair in a Morse chart (they are all homotopic anyway), collapse the stable part and quotient by the exit set. By doing this we get the commutative diagram:

$$\begin{array}{ccc}
 K^{k-1}(B, C) & \xrightarrow{\delta_{triple}} & K^k(A, B) \\
 \downarrow & & \downarrow \\
 K^{k-1}(S^{k-1}) & \longrightarrow & K^k(S^{k-1}) \\
 \downarrow & \nearrow & \\
 K^k(S^k) & &
 \end{array}$$

And as a map between K Groups of spheres, we know that this map corresponds to the question of orientation. Now we have the maps

- (a) $S^k \rightarrow S \vee (B/C)$ induced from an orientation in the unstable manifold around p together with $-\text{grad}(f)(x_i)$ where $x_i \in W(q \rightarrow p)$.
- (b) $S^k \rightarrow (A/B)$ induced from an orientation in the unstable manifold around q .

And hence the diagram

$$\begin{array}{ccc} K^k(S \vee (B, C)) & \xrightarrow{\delta_{\text{triple}}} & K^k(A, B) \\ & \nwarrow \quad \nearrow & \\ & K^k(S^k) & \end{array}$$

The map now factors over The K-Group of a bouquet of k -spheres generated over the connecting flow lines. In each such sphere the comparison of orientations happens.

Before we start proving this we need the lemma:

Lemma 4.1.11. *We work in a compact, riemanian, orientable, smooth Manifold (M, g) of dimension m together with a Morse-Smale funktion*

$$f : M \rightarrow \mathbb{R} \tag{9}$$

We assume that q is a critical point of index $k+1$ and p is a critical point of index k . we define $f(q) = b$ and $f(p) = a$ and assume that there is no kritical point in $f^{-1}(a, b) \subseteq M$. We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \leq t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \geq t\}, \tag{10}$$

and the constants

$$c \in (a, b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big} . \tag{11}$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \leq b + \varepsilon\}, \tag{12}$$

$$L_q := \{x \in N_q | f(x) = c\}, \tag{13}$$

$$N_p := \{x \in M^c | f(\varphi_T(x)) \geq a - \varepsilon\}, \tag{14}$$

$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}, \tag{15}$$

and finally:

$$C := N_p \cup N_q, \quad (16)$$

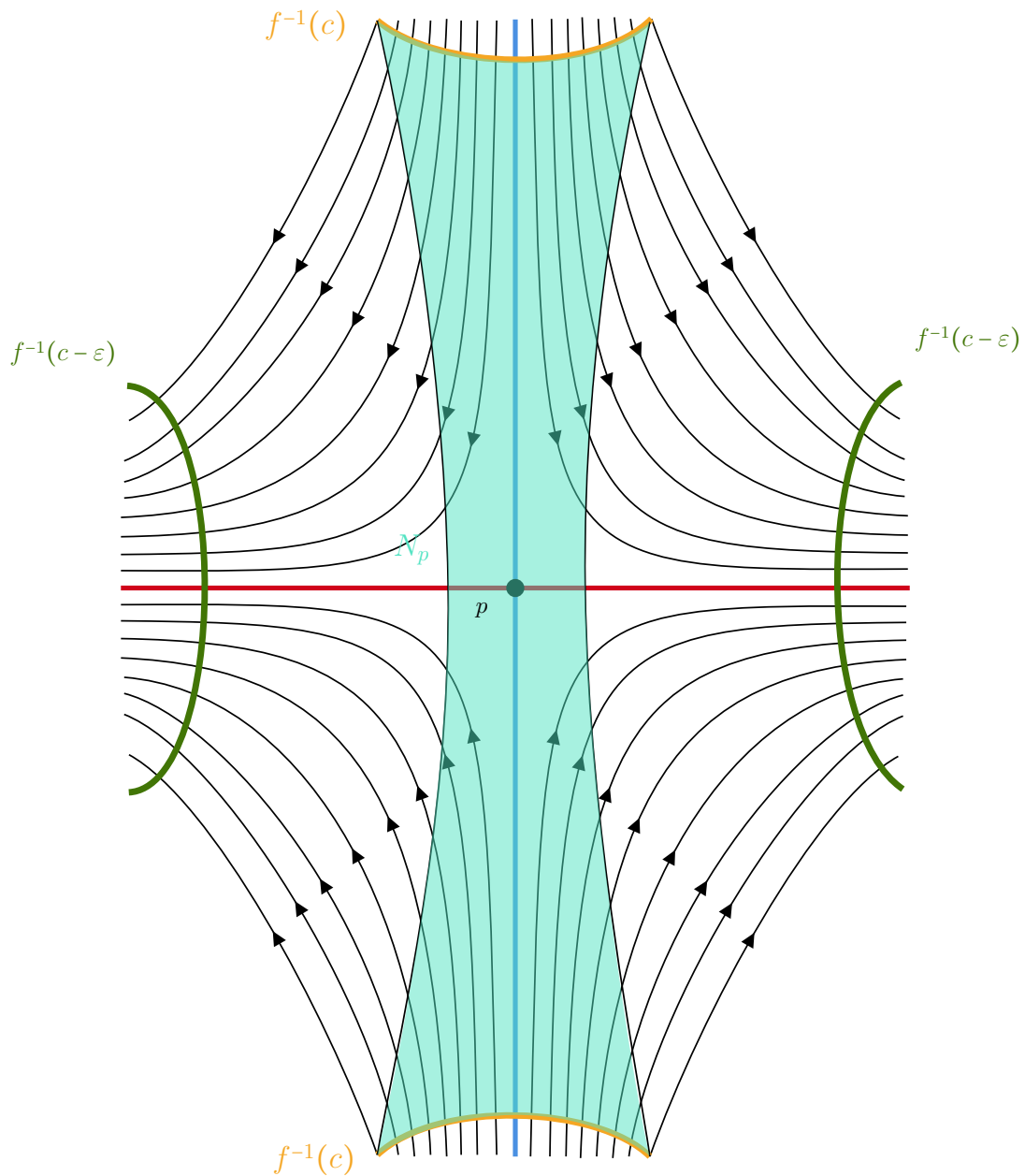
$$B := N_p \cup L_q, \quad (17)$$

$$A := L_p \cup (L_q - \overset{\circ}{N}_p). \quad (18)$$

We claim that

1. (N_q, L_q) is a regular index pair for q .
2. (C, B) is an index pair for q .
3. (N_p, L_p) is a regular index pair for p .
4. (B, A) is an index pair for p .
5. N_p is homotopically equivalent to a tubular neighbourhood of $W(\rightarrow p) \cap M^c$.

The statements 1-4 are easy to proof by the definitions. The regularity can be derived from a general fact for pairs (X, Y) in metric spaces: If Y is closed in X and there is a neighbourhood of Y that is open in X such that Y is a strong deformation retract of U . So the only thing left to show is, that N_p is a tubular neighbourhood. in [MorseTheorySalmon] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [banyaga2004lectures] Banyaga claims this (also without any argument). I find this difficult to proof, since we cannot use the flow for the map from the normal bundle, as points leave N_p along the flow: The Set N_p is sketched in the figure below.



So to prove this, we first formulate two lemmas. First we restate/ refine the λ -lemma:

Lemma 4.1.12 (The Technical λ -Lemma). *Let M be a smooth manifold and let p be a hyperbolic fixpoint of a diffeomorphism φ on M with $\dim W(p \rightarrow) = r$ for $0 < r < m = \dim M$. Let N be an injective immersed submanifold of M that is invariant under φ and has a point of transverse intersection q with $W(\rightarrow p)$. Then for any $\epsilon > 0$, any r -cell neighbourhood D of q in N that also intersects*

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$W(\rightarrow p)$ transversal and any r -cell neighbourhood B of p in $W(p \rightarrow)$ there is a smaller cell $\tilde{D} \subset D$, an $n \in \mathbb{N}$ such that the following is true:

We can assume that B and $\varphi^n(\tilde{D})$ lives in a chart centered at p with a product structure $U \cong W(\rightarrow p) \cap B_\mu(0) \times W(p \rightarrow) \cap B_\mu(0)$. Here, the inverse of the projection into $B \subseteq W(\rightarrow p)$:

$$h : B \rightarrow \varphi^n(\tilde{D})$$

satisfies:

1. $\sup_{x \in B} \|x - h(x)\| < \varepsilon$ and
2. For the inclusions $i_1 : B \rightarrow M$ and $i_2 : \varphi^n(\tilde{D}) \rightarrow M$ we have that

$$\sup_{x \in B} \|d(i_1)_x - d(i_2 \circ h)_x\| < \varepsilon.$$

Proof. The proof is essentially the same as given in David Hurtubise - Lectures on Morse Homology. I omitted it here. $\square$

Next we prove that fiberwise the flow is monotonically decreasing in any normal-direction with respect to $W(\rightarrow p)$:

Lemma 4.1.13. *Let U be a tubular neighbourhood of $W(\rightarrow p) \cap M^c$, with $J : NW(\rightarrow p) \cap M^c \rightarrow M$ being the corresponding embedding. For $x \in W(\rightarrow p) \cap M^c$ we denote U_x the embedded fiber over x . Then there exists a $T > 0$ big enough and a global trivialization $\Phi : NW(\rightarrow p) \cap M^c \rightarrow (W(\rightarrow p) \cap M^c) \times \mathbb{R}^{\lambda_p}$ such that the composition $f \circ \varphi_T \circ J \circ \Phi^{-1}$ is fibrewise monotonically decreasing meaning that for any $0 \neq v \in U_x$ and $\nu > 0$ small the function*

$$l_\nu : (1 - \nu, 1 + \nu) \rightarrow \mathbb{R}$$

$$\lambda \mapsto f \circ \varphi_T \circ J \circ \Phi^{-1}(x, \lambda \cdot v)$$

has negative derivative.

Proof. We will apply the technical λ -lemma. For this we reuse the notation as follows. The lemma tells us, that for any $\mu_T > 0$ we can find a T big enough, such that $\varphi_T(J(U_x))$ is close in the following sense: There is a neighbourhood L of p and a Morse chart

$$\Psi : L \rightarrow \mathbb{R}^{\lambda_p} \times \mathbb{R}^{m-\lambda_p}$$

such that $\Psi|_{\mathbb{R}^{\lambda_p}} \supseteq W(p \rightarrow)$. In this chart the set $\varphi_T(J(U_x))$ is the graph of a function $h_x^s : \mathbb{R}^{\lambda_p} \rightarrow \mathbb{R}^{m-\lambda_p}$ that satisfies:

$$\|dh_v^s\| < \mu_T$$

$$\|h_v^s\| < \mu_T$$

We define the map $h_x = h_x^s \oplus \text{id}_{\lambda_p} : \mathbb{R}^{\lambda_p} \rightarrow \mathbb{R}^m$.

Now $\varphi_T(J(U_x)) = \Gamma(h_x^s) = \text{im}(h_x)$. Using this we define our global trivialisation:

$$\begin{aligned} \Phi : NW(\rightarrow p) \cap M^c &\rightarrow (W(\rightarrow p) \cap M^c) \times \mathbb{R}^{\lambda_p} \\ \xi &\mapsto \left(\pi(\xi), h_{\pi(\xi)}^{-1} \circ \Psi \circ \varphi_T \circ J(\xi) \right) \end{aligned}$$

In this trivialization the funktion l_v looks like:

$$\begin{aligned} l_v(\lambda) &= f \circ \varphi_T \circ J \circ \Phi^{-1}(x, \lambda v) \\ &= f \circ \varphi_T \circ J \circ J^{-1} \circ \varphi_{-T} \circ \Psi^{-1} \circ h_x \\ &= f \circ \Psi^{-1} \circ h_x. \end{aligned}$$

In coordinates this reads:

Now we can calculate its derivative to conclude the statement:

$$\begin{aligned} d(l_v)(\lambda) &= d(f \circ \psi^{-1})|_{h_x(\lambda v)} \circ dh_x|_{\lambda v}(v) \\ &= d(f \circ \psi^{-1})|_{h_x(\lambda v)}(v, dh_x^s(v)) \\ &= -2\lambda \sum_{i_1}^{\lambda_p} (v^{i_1})^2 + (h_x^s(\lambda v))^T \cdot (dh_x^s)^i(\lambda v) \\ &\leq -2\lambda \|v\|^2 + \lambda^2 (\mu^T)^2 \|v\|^2 \\ &= (-2 + (\mu^T)^2 \lambda) \lambda \|v\|^2 < 0 \text{ for all } v \neq 0, \lambda \in I \text{ and for all } T \text{ big enough.} \end{aligned}$$

□

Hence, the sets $N_p \cap U_x$ are starshaped with respect to the embedding of the normal space.

Now we want to **smoothly** rescale to the starshaped neighbourhood. This is in general not possible and homeomorphisms from starshaped neighbourhoods to R^n are hard to describe, even if they are not required to be a rescaling. The main problem in the general case is, that the following: for each $v \in \varphi_T(J(U_x))$ by the star shaped property there is a unique λ_v , such that $\lambda_v v$ such that $f \circ (\lambda_v v) = a - \varepsilon$. But the map $v \mapsto \lambda_v$ needs not to be continuous in the general starshaped setting however here it is and it even continuously depends on x .

Definition 4.1.14 (Hyperspherical coordinates). For our next proof we need to intro-

duces spherical coordinates on R^{λ_p} . Define $s_k := \sin(\theta_k)$ and $c_k := \cos(\theta_k)$:

$$\Lambda : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow \mathbb{R}^{\lambda_p}$$

$$(r, \theta_1, \dots, \theta_{\lambda_p-1}) \mapsto \begin{pmatrix} rc_1 \\ rs_1c_2 \\ \vdots \\ rs_1 \cdots s_{\lambda_p-2}c_{\lambda_p-1} \\ rs_1 \cdots s_{\lambda_p-2}s_{\lambda_p-1} \end{pmatrix}$$

With differential

$$\begin{pmatrix} c_1 & -rs_1 & 0 & 0 & \cdots & 0 \\ s_1c_2 & rc_1c_2 & -rs_1s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ s_1 \cdots s_{n-2}c_{n-1} & \cdots & \cdots & \cdots & -rs_1 \cdots s_{n-2}s_{n-1} \\ s_1 \cdots s_{n-2}s_{n-1} & rc_1 \cdots s_{n-1} & \cdots & \cdots & rs_1 \cdots s_{n-2}c_{n-1} \end{pmatrix}$$

We have shown in lemma 4.1.13 that the maps l_v are monotonically decreasing. Hence the condition $f(x) \geq a - \varepsilon$ gives rise to a starshaped fiber. But we want the fibers to be homeomorphic to R^{λ_p} and the dream would be to just rescale it as follows: Assume that $\rho > 0$ is small enough, such that for all $x \in (W(\rightarrow p) \cap M^c)$ we have that

$\sup_{x \in h_x(B_\rho(0))} (f(x)) \geq a - \varepsilon$. Now we have the maps:

$$r_x : h_x(\partial(B_\rho(0))) \rightarrow \mathbb{R}_{\geq 1}$$

$$v \mapsto \lambda_v \text{ such that } f(\lambda_v v) = a - \varepsilon$$

Notice that for small enough ε such a λ_v exists and the lemma 4.1.13 gives us uniqueness. Now we want to have continuity and furthermore continuous dependence on x for which we will use the parameter-dependent implicit function theorem:

Lemma 4.1.15. *The parameterdependent functions r_x are continuous and depend continuous on x*

Proof. We redefine the funktion r_x as follows: Implizitly for each x we have the Mapping:

$$F_x : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow \mathbb{R}$$

$$(r, \theta_1, \dots, \theta_{\lambda_p-1}) \mapsto f \circ h_x \circ \Lambda(r, \theta_1, \dots, \theta_{\lambda_p-1})$$

f mit $f \circ \Psi^{-1}$ ersetzen, da hier eigentlich die Morse karte genutzt wird!!

Here the funktion reads:

$$f \circ h_x \circ \Lambda(\underbrace{r, \theta_1, \dots, \theta_{\lambda_p-1}}_{:=v}) = f\left(\begin{pmatrix} h_x^s(\Lambda(v)) \\ \Lambda(v) \end{pmatrix}\right) = -2r^2 + 2 \sum_{i=\lambda_p+1}^m (h_x(\Lambda(v)))_i^2$$

To make use of the implizit funktion theorem, we inspect the differential in r -direktion:

$$\frac{\partial}{\partial r}(-2r^2 + 2 \sum_{i=\lambda_p+1}^m (h_x(\Lambda(v)))_i^2) = -4r + 2 \sum_{i=\lambda_p+1}^m \left(\sum_{j=1}^{\lambda_p} \left(\frac{\partial(h_x)_i}{\partial x_j} \frac{\partial \Lambda_j}{\partial r} \right) \right)$$

This does not vanish for big enough r , since the partial differentials of λ are of norm one, and those of h_x are small. Hence for big enough r , lets say $r \geq \mu > 0$ this is negative and thereby we get a funktion

$$\tilde{r}_x : [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow \mathbb{R}_{\geq \mu}$$

such that $(f \circ h_x \circ \Lambda)^{-1}(a - \varepsilon)$ is the graph of $\tilde{r}_x$. Hence, if we define $\pi_r : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p-2} \times [0, 2\pi) \rightarrow [0, \pi]^{\lambda_p-2} \times [0, 2\pi)$ to be the projection we have that

$$\begin{aligned} r_x : h_x(\partial B_\rho(0)) &\rightarrow \mathbb{R} \\ v &\mapsto \tilde{r}_x \circ \pi_r(\Lambda^{-1}(h_x^{-1}v)) \end{aligned}$$

is continouos and depends continouosly on x . (In reality we still need to check for smoothness in $\Lambda(\{\theta_{\lambda_p-1} = 0\})$. This can be done by a different parametrisation of the sphere meaning by different spherical coordinates.) The resulting funktion agrees with the earlier definition of r_x concluding the proof. $\square$

Proof That N_p is Tubular. To finish our proof we rescale the Fibers smoothly. For this we first restrict the domain of J : Let $\nu > 0$ be small enough, such that for

$$N_\nu W(\rightarrow p) \cap M^c := \{\xi \mid \text{if } \Phi(\xi) = (x, v), \|v\| < \nu\}$$

the supremum

$$\sup_{x \in \varphi_T \circ J(N_\nu W(\rightarrow p) \cap M^c)} \geq a - \varepsilon$$

Hence, $J(N_\nu W(\rightarrow p) \cap M^c)$ is included in N_p and now we resscale the fibres $U_x^\nu := U_x \cap \text{im} J(N_\nu W(\rightarrow p) \cap M^c) \supseteq U_x$ to get a surjective inclusion into N_p . For all $x \in W(\rightarrow p) \cap M^c$ we define the smooth rescaling as follows:

$$\begin{aligned} RS : U_x^\nu &\rightarrow U_x \\ v &\mapsto \varphi_{-T} \circ \Psi^{-1} \circ r_x \left(\rho \cdot \frac{h_x^{-1} \circ \Psi \circ \varphi_T(v)}{\|h_x^{-1} \circ \Psi \circ \varphi_T(v)\|} \right) \end{aligned}$$

This rescaling is smooth, smoothly depends on x , and it maps surjectifly onto $N_p \cap U_x$. $\square$

Proof of Theorem 4.1.10. Lets say that $p \in -\mathbb{N}$ is negative for the moment. By definition we get the triple connecting morphism as the composition of the inclusion with the connecting homomorphism of the pair:

$$\begin{array}{ccccc}
 & & K^p[N_q \cup N_p, L_q \cup N_p] & \xlongequal{\quad} & K^p[N_q \cup N_p \cup (L_q \cup N_p)] \\
 & \nearrow \delta_{\text{pair}} & & & \uparrow (m^*)^{-1} \\
 & & & & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p)] \\
 & & & & \uparrow i_\alpha^* \\
 K^{p-1}[L_q \cup N_p] & \xlongequal{\quad} & K^p[S_1 \wedge (L_q \cup N_p)] & \xrightarrow{\alpha^*} & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \\
 \uparrow i^* & & \uparrow i_\beta^* & & \\
 K^{p-1}[L_q \cup N_p, L_p \cup (L_q \setminus \dot{N}_p)] & \xlongequal{\quad} & K^p[S_1 \wedge L_q \cup N_p, S_1 \wedge L_p \cup (L_q \setminus \dot{N}_p)] & &
 \end{array}$$

The bottom left diagramm commutes by definition. All maps with names $i_{\text{something}}^*$ are induced from obvious inclusions. The other maps are defined as follows, where p denotes the choosen point of our pointed spaces:

$$\begin{aligned}
 \alpha : N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p) &\rightarrow S_1 \wedge (L_q \cup N_p) \\
 x &\mapsto \begin{cases} p & \text{if } x \in C_2(N_q \cup N_p) \\ x & \text{else} \end{cases}
 \end{aligned}$$

This map is well defined and continouos, as every point in $S_1 \wedge (L_q \cup N_p)$ is of the form $[(t_1, x)]$ where $x \in L_q \cup N_p$. and those are the points in the domain of α that not get collapsed to p We have the maps:

$$\begin{aligned}
 c_\gamma : [N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] &\rightarrow \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \middle/ \frac{C_1(L_p \cup (L_q \setminus \dot{N}_p)) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right] \\
 x &\mapsto \begin{cases} p & \text{if } x \in C_2(N_q \cup N_p) \cup C_1(L_p \cup (L_q \setminus \dot{N}_p)) \\ x & \text{else} \end{cases}
 \end{aligned}$$

and

$$\begin{aligned}
 c_\beta : [S_1 \wedge (L_q \cup N_p)] &\rightarrow \left[S_1 \wedge L_q \cup N_p \middle/ S_1 \wedge L_p \cup (L_q \setminus \dot{N}_p) \right] \\
 x &\mapsto \begin{cases} p & \text{if } x \in S_1 \wedge L_p \cup (L_q \setminus \dot{N}_p) \\ x & \text{else} \end{cases}
 \end{aligned}$$

This map is again well defined and continouos and is also a map of pairs. In fact, both maps α and β can be written as $x \mapsto \bar{x}$ and since the collapsed space is contractible, they induce isomorphism between the K -groups. The map γ is just the identity.

Furthermore, we have the commutative diagramm:

$$\begin{array}{ccc}
 K^p[S_1 \wedge (L_q \cup N_p)] & \xrightarrow{\alpha^*} & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \\
 \uparrow c_\beta^* & & \uparrow c_\gamma^* \\
 K^p[S_1 \wedge L_q \cup N_p \middle/ S_1 \wedge L_p \cup (L_q \setminus \dot{N}_p)] & \xrightarrow{\gamma^*} & K^p \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \middle/ \frac{C_1(L_p \cup (L_q \setminus \dot{N}_p)) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right]
 \end{array}$$

To see the commutativity we have to show that

$$c_\beta \circ \alpha = \gamma \circ c_\gamma : N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p) \rightarrow \left[S_1 \wedge L_q \cup N_p / S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \right]$$

since α and β are both of the form $x \mapsto \bar{x} = x$ or p we compare the case distinctions:

- $c_\beta \circ \alpha(x) = p \Leftrightarrow x \in C_1(L_p \cup (L_q \setminus \mathring{N}_p)) \cup C_2(N_q \cup N_p)$
- $\gamma \circ c_\gamma(x) = p \Leftrightarrow x \in C_2(N_q \cup N_p) \cup C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)$

Which are the same, since $L_q \setminus \mathring{N}_p = \overline{L_q \setminus N_p}$ due to L_q being closed. Now we consider the maps:

$$c : [N_q \cup N_p \cup C_1(L_q \cup N_p)] \rightarrow \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p)}{\text{cl} \left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^l C_1 V_j \right)} \right]$$

$$x \mapsto \begin{cases} p & \text{if } x \in \text{cl} \left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^l C_1 V_j \right) \\ x & \text{else} \end{cases}$$

and

$$c_i : [N_q \cup N_p \cup C_1(L_q \cup N_p)] \rightarrow \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} / \frac{N_q \cup N_p \cup C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right]$$

$$x \mapsto \begin{cases} p & \text{if } x \in N_q \cup N_p \cup C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p) \\ x & \text{else} \end{cases}.$$

We define V_j to be the finitely many components $N_p \cap S(q \rightarrow)$ and hence we can identify the sets $\overline{L_q \setminus N_p} = \text{cl}(L_q \setminus (L_q \cap N_q)) = \text{cl}(L_q \setminus (\bigcup_{j=1}^l V_j))$ and with this the map η in the diagram below is just the identity:

$$\begin{array}{ccc}
& K^p[N_q \cup N_p \cup (L_q \cup N_p)] & \\
& \uparrow (m^*)^{-1} & \\
K^p[N_q \cup N_p \cup C_1(L_q \cup N_p)] & \xleftarrow{c^*} & K^p \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p)}{\text{cl} \left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^l C_1 V_j \right)} \right] \\
& \uparrow i_\alpha^* & \uparrow i_\delta^* \\
K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] & & K^p \left[\frac{C_1(L_q \cup N_p)}{\text{cl} \left(C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^l C_1 V_j \right)} \right] \\
& \uparrow c_\gamma^* & \uparrow i_\eta^* \\
& & K^p \left[\frac{C_1(L_q)}{\text{cl} \left(C_1(L_q) \setminus \bigcup_{j=1}^l C_1 V_j \right)} \right] \\
& & \uparrow c_\epsilon^* \\
K^p \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right] & \xleftarrow{\eta^*} & K^p \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right] \Big/ \frac{N_q \cup N_p \cup C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)}
\end{array}$$

Now the mammoth task is to show that this beast commutes: For this we proceed with the same strategy since again all maps are inclusions and collapses and hence the maps look like $x \mapsto \bar{x} = x$ or p . For the consideration assume that $x \neq p$ in the domain. Then:

$$\eta \circ c_\gamma \circ i_\alpha(x) = p \Leftrightarrow x \in C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)$$

On the other hand:

$$\begin{aligned}
c_\epsilon \circ i_\eta \circ i_\delta \circ c(x) = p &\Leftrightarrow c(x) = p \text{ or } c_\epsilon(x) = p \\
x \in \text{cl} \left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^l C_1 V_j \right) \cup N_q \cup N_p \cup C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p) \\
&= C_1 \text{cl} \left((L_q \cup N_p) \setminus \bigcup_{j=1}^l V_j \right) \cup C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p) \\
&= C_1 \text{cl} \left((L_q) \setminus \bigcup_{j=1}^l V_j \right) \cup C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p) \\
&= C_1(L_p \cup \text{cl} \left(L_q \setminus \bigcup_{j=1}^l V_j \right)) \cup C_2(N_q \cup N_p)
\end{aligned}$$

And now the conditions agree and hence the big diagram commutes. Now notice how we have the homeomorphism

$$\Lambda : \left[\frac{C_1(L_q)}{\text{cl} \left(C_1(L_q) \setminus \bigcup_{j=1}^l C_1 V_j \right)} \right] \rightarrow \bigvee_{j=1}^l C_1 V_j / \partial C_1 V_j$$

that is the identity on all $C_1 \mathring{V}_j$. Furthermore we have the compression maps

$$c_j : \bigvee_{j=1}^l C_1 V_j / \partial C_1 V_j \rightarrow C_1 V_j / \partial C_1 V_j$$

Hence by the additivity of K -Theorie, the pair

$$\left(K^p \left[\frac{C_1(L_q)}{\text{cl} \left(C_1(L_q) \setminus \bigcup_{j=1}^l C_1 V_j \right)} \right], ((c_j \circ \Lambda)^*)_j \right)$$

satisfies the coproduct universal property.

So consider the map

$$\kappa : c^* \circ i_\delta^* \circ i_\eta^* \circ (c_j \circ \Lambda)^* : K^p [C_1 V_j / \partial C_1 V_j] \rightarrow K^p [N_q \cup N_p \cup C_1(L_q \cup N_p)] \quad (19)$$

This map is induced from

$$\begin{aligned} N_q \cup N_p \cup C_1(L_q \cup N_p) &\rightarrow C_1 V_j / \partial C_1 V_j \\ x &\mapsto \begin{cases} x & \text{if } x \in C_1 \mathring{V}_j, \\ p & \text{else.} \end{cases} \end{aligned}$$

For all $C_1 V_j$ we need an homeomorphism to S^{k+1} :

$$\begin{aligned} r : C_1 V_j / \partial C_1 V_j &\rightarrow I \times D^k / \partial(I \times D^k) \\ (t, x) &\mapsto (t, \psi(x)) \end{aligned}$$

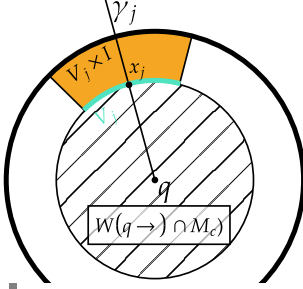
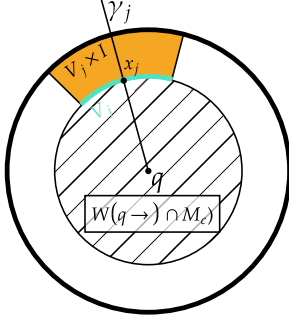
For this we use that N_p is a tubular neighbourhood of $W(\rightarrow p) \cap M^c$ where that latter is contractable. Hence, we have a homotopy equivalence by contracting the base space and the fibers simultaneously via a frame. and the transversality ensures, that this gives a homeomorphism from V_j to the fibre over p , which is a ball $B \subseteq W(p \rightarrow)$. and composed with the refined Morse chart this is a Ball B^{λ_p} in $\mathbb{R}^{\lambda_p}$. This gives us a well defined homeomorphism:

$$r : C_1 V_j / \partial(C_1 V_j) \rightarrow C_1 B^{\lambda_p} / \partial(C_1 B^{\lambda_p})$$

Hence, an orientation on $TW(p \rightarrow)$, induces a generator α of $K^p [C_1 V_j / \partial(C_1 V_j)]$. **Here the orientation is introduced by choosing the refined Morse chart.**

On the other hand, we have a generator on $N_p \cup N_p \cup C_1(L_p \cup N_p)$ induced from an orientation on $TW(q \rightarrow)$ as follows: First we get a generator of the K group of the Regular index pair $(N_q/L_q) \simeq (N_q \cup N_p/L_q \cup N_p)$ by choosing a small index pair and the homotopy. This generator induces a generator β of $K^p [N_q \cup N_p \cup C_1(L_q \cup N_p)]$ by pulling it back via the contraction of the cone C_1 . Now we need to study, how the image of α under the map (19) relates to β :

Hier ist noch einiges zu tun!



For this we make use of an collar neighbourhood of $W(q \rightarrow) \cap M_c$ defined as follows: Again using the implicit function theorem and spherical coordinates (compare [referenz](#)) we have that the preimage of $W(q \rightarrow) \cap M_c$ under the embedding $E_q^u : T_q^u M \rightarrow W(q \rightarrow)$ is (after rescaling the function) a Disk, So without restrictions assume that $E_p^u(B_1(0)) = W(q \rightarrow) \cap M_c$. Then $\mathfrak{C} := E_p^u(B_2(0))$ is a collar neighbourhood and $V_j \times I \hookrightarrow L_i \subseteq \mathfrak{C}$ embeds in the obvious way. This is done by mapping $V_j \times I$ first into $B_2(0)$ via $(v, t) \mapsto (E_p^u)^{-1}v \cdot (1 + t)$ and then applying E_p^u .

Furthermore, $V_j \times I$ is homeomorphic to $\mathfrak{C}$ by stretching it. The stretching can be done along rays starting in $y_j = (x_j, 1)$. This is homotopic to the identity, if viewed as an endomorphism in $W(q \rightarrow) \cap M_c$. We can now define the collapsing map $W(q \rightarrow) \cap M_c$ in a continuous way by defining it to be the identity inside of L_j and outside we map each point to the intersection of the ray containing said point with the boundary of L_p .

Proof Strategy:

The idea of the next proof is the following: an orientation in $N_{x_j}\gamma_j$ induces a generator of the k -th K group $C_1V_j/\partial(C_1v_j)$. This generator can be pulled back to be a generator of the k -th K group of $W(q \rightarrow) \cap M_c/\partial(W(q \rightarrow) \cap M_c)$ via the collapsing map. This pulling back coincides with the map from (19). Now assume that the basis $(b_j, -\text{grad}(f))$ of $N_{x_j}\gamma_j \oplus T_{x_j}\gamma_j$ is positively oriented with respect to the orientation of $T_q^u M$. And assume furthermore, that the inverse of the embedding E_q^u is an oriented chart. Then the pulled back generator agrees with the generator of $W(q \rightarrow) \cap M_c/\partial(W(q \rightarrow) \cap M_c)$ that is induced by directly using E_q^u .

□